Q - Farming Fish.

In a fish farm, fish are harvested at a constant rate of 2100 fish per week. The per capita death rate for the fish is 0.2 fish per day per fish and the per capita death rate is 0.7 fish per day per fish.

a) Write down a word equation describing the rate of change of the fish population. Hence obtain a differential equation for the number of fish. (Define any symbols you introduce)

b) Find the general solution of the differential equation obtained in part (a).

c) Determine if there are any values for which the fish population is in equilibrium.

Solution: Let $N(t)$ = no. of fish in the farm at time t

where t is measured in days

Harvesting rate = 2100 fish per week

$= \frac{2100}{7} = 300$ fish per day

per-capita birth rate = 0.7 fish / day / fish

$\Rightarrow$ Total birth per day = $0.7N$ per day

per-capita death rate = 0.2 fish / day / fish

$\Rightarrow$ Total death per day = $0.2N$ per day

a) word equation

Rate of change of fish population

= birth rate - death rate - harvesting rate

• Differential Eq^{n}

$$\frac{dN}{dt} = 0.7N - 0.2N - 300$$

$$\boxed{\frac{dN}{dt} = 0.5N - 300}$$

b) General Solution

$$\frac{dN}{dt} - 0.5N = -300$$

This is linear differential equation

$$I.f = e^{-\int 0.5\,dt} = e^{-0.5t}$$

$$\Rightarrow N e^{-0.5t} = \int (-300)\, e^{-0.5t}\, dt$$

$$N e^{-0.5t} = 600\, e^{-0.5t} + C$$

$$\boxed{N = 600 + C e^{0.5t}}$$

where C is arbitrary constant.

c) For the equilibrium pt.

$$\frac{dN}{dt} = 0$$

$$\Rightarrow 0.5N - 300 = 0$$

$$\boxed{N = 600}$$

(b) Can also do ~~with~~ using Separation of Variables.

$$\frac{dN}{dt} = .5N - 300$$

$$\Rightarrow \frac{dN}{.5N-300} = dt \Rightarrow \int \frac{dN}{.5N-300} = \int dt$$

$$\frac{1}{.5} \ln(.5N-300) = t + C_1$$

$$\Rightarrow \ln(.5N-300) = .5t + .5C_1$$

$$\Rightarrow \ln(.5N-300) = .5t + .5C_1 \quad \text{(optional)}$$

$$\Rightarrow .5N = 300 + C_2 e^{.5t} \quad (.5C_1 = C_2)$$

$$\Rightarrow N = \frac{300}{.5} + \frac{C_2}{.5} e^{.5t} \quad \text{(optional)}$$

$$\Rightarrow \boxed{N = 600 + C e^{.5t}} \quad \left(C = \frac{C_2}{.5}\right)$$

Question 2:

Sol^{n}: Let y_1 and y_2 are any two Sol^{n}s of the differential equation

$$a_0(x)\,y'' + a_1(x)\,y' + a_2(x)\,y = 0, \quad \forall x \in [a,b]$$

where $a_0(x) \neq 0$, $a_1(x)$, $a_2(x)$ are Constant $\forall x \in [a,b]$.

Then

$$w(y_1,y_2)(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$= y_1 y_2' - y_2 y_1'$$

So, $w'(y_1,y_2)(x) = \cancel{y_1' y_2'} + y_1 y_2'' - \cancel{y_2' y_1'} - y_2 y_1''$

$$= y_1 y_2'' - y_2 y_1''$$

Now; $a_0(x)\,y_1'' + a_1(x)\,y_1' + a_2(x)\,y_1 = 0$ $\quad (\times y_2)$

$a_0(x)\,y_2'' + a_1(x)\,y_2' + a_2(x)\,y_2 = 0$ $\quad (\times y_1)$

on substracting we get;

$$\Rightarrow a_0(x)\left(y_1'' y_2 - y_2'' y_1\right) + a_1(x)\left(y_2 y_1' - y_1 y_2'\right) = 0$$

$$\Rightarrow -a_0(x) w' - a_1(x) w = 0$$

$$\Rightarrow a_0(x) w' + a_1(x) w = 0$$

$$\Rightarrow \frac{w'}{w} = -\frac{a_1(x)}{a_0(x)}$$

$$\Rightarrow \underbrace{\log w(x)}_{(\text{or } \ln(w(x)))} = -\int \frac{a_1(x)}{a_0(x)} dx + C$$

$$\Rightarrow w(x) = c\, e^{-\int \frac{a_1(x)}{a_0(x)} dx}.$$

Note: Some people can write

$$\ln|w(x)| = -\int \frac{a_1(x)}{a_0(x)} dx + C_1$$

$$\text{so, } |w(x)| = c\, e^{-\int \frac{a_1(x)}{a_0(x)} dx}.$$

Some people after this can write:

$$w(x) = e^{-\int \frac{a_0(x)}{a_1(x)} dx}$$

(underlying assumption is either the constant is C or may change to $-C$)

Question 3:

Sol^{n}: The corresponding homogeneous equation is

$$y'' - 3y' + 2y = 0$$

The characteristic eq^{n} is $\lambda^2 - 3\lambda + 2 = 0$

$$\Rightarrow (\lambda - 2)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = 1, 2.$$

So, the complementary f^{n} / general sol^{n} of the homogeneous eq^{n} is

y_c or y_H $\Rightarrow y = c_1 e^{x} + c_2 e^{2x}$

Thus we take as our particular solution

$$y_p = Ax^2 + Bx + C + De^{3x} + Ex^2e^x + Fxe^x$$

For this, we have

$$y_p' = 2Ax + B + 3De^{3x} + Ex^2e^x + 2Exe^x + Fxe^x + Fe^x,$$

$$y_p'' = 2A + 9De^{3x} + Ex^2e^x + 4Ee^x + Fxe^x + 2Fe^x$$

We substitute y_p, y_p' and y_p'' into the differential equation for y, y' and y'', respectively to obtain;

$$2A + 9De^{3x} + Ex^2e^x + (4E+F)xe^x + (2E+2F)e^x - 3[2Ax + B + 3De^{3x} + Ex^2e^x + (2E+F)xe^x + Fe^x] + 2(Ax^2 + Bx + C + De^{3x} + Ex^2e^x + Fxe^x)$$

$$= 2x^2 + e^x + 2xe^x + 4e^{3x}.$$

OR

$$(2A-3B+2C)+(2B-6A)x+$$
$$2Ax^2+2De^{3x}+(-2E)xe^x+(2E-F)e^x$$
$$=2x^2+e^x+2xe^x+4e^{3x}$$

Equating coefficients of like terms, we have:

$$2A-3B+2C=0$$
$$2B-6A=0$$
$$2A=2$$
$$2D=4$$
$$-2E=2$$
$$2E-F=1$$

From this $A=1$, $B=3$, $C=\frac{7}{2}$, $D=2$, $E=-1$, $F=-3$, and so the particular integral is

$$y_p=x^2+3x+\frac{7}{2}+2e^{3x}-x^2e^x-3xe^x$$

The general Soln is therefore

$$y=y_c+y_p$$
$$=c_1e^x+c_2e^{2x}+x^2+3x+\frac{7}{2}+2e^{3x}-x^2e^x-3xe^x.$$

□

This is the alternative Soln of 2nd part of problem 3:

(only the particular Soln)

(3) The particular solution using method of Variation of parameter.

Method of Variation of Parameter.

$y_1 = e^x$, $y_2 = e^{2x}$

$$W(y_1, y_2) = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{3x} - e^{3x} = e^{3x} \neq 0$$

$$r(x) = 2x^2 + e^x + 2xe^x + 4e^{3x}$$

$$\int \frac{r y_2}{W} dx = \int \frac{(2x^2 + e^x + 2xe^x + 4e^{3x})}{e^{3x}} e^{2x} dx$$

$$= \int (2x^2 + e^x + 2xe^x + 4e^{3x}) e^{-x} dx$$

$$= 2\int x^2 e^{-x} dx + \int dx + \int 2x dx + 4\int e^{2x} dx$$

$$= 2\left[-x^2 e^{-x} - \int 2x \frac{e^{-x}}{(-1)} dx \right] + x + 2\frac{x^2}{2} + 4\frac{e^{2x}}{2}$$

$$= 2\left[-x^2 e^{-x} + 2\left\{ x\frac{e^{-x}}{(-1)} - \int \frac{e^{-x}}{(-1)} dx \right\} \right] + x + x^2 + 2e^{2x}$$

$$= 2\left[-x^2 e^{-x} - 2xe^{-x} - 2e^{-x} \right] + x + x^2 + 2e^{2x}$$

$$= -2x^2 e^{-x} - 4xe^{-x} - 4e^{-x} + x + x^2 + 2e^{2x}$$

$$\int \frac{r y_1}{W}\,dx = \int \frac{(2x^2+e^x+2xe^x+4e^{3x})}{e^{3x}} e^x\,dx$$

$$= \int (2x^2+e^x+2xe^x+4e^{3x})\,e^{-2x}\,dx$$

$$= 2\int x^2 e^{-2x}\,dx + \int e^{-x}\,dx + 2\int x e^{-x}\,dx + 4\int e^x\,dx$$

$$= 2\left[-x^2\frac{e^{-2x}}{2} - \int 2x\,\frac{e^{-2x}}{(-2)}\,dx\right] - e^{-x} + 2\left[x\frac{e^{-x}}{(-1)} - \int 1.\frac{e^{-x}}{(-1)}\,dx\right] + 4e^x$$

$$= 2\left[-\frac{x^2e^{-2x}}{2} + x\frac{e^{-2x}}{(-2)} - \int 1.\frac{e^{-2x}}{(-2)}\,dx\right] - e^{-x} + 2\left[-xe^{-x} - e^{-x}\right] + 4e^x$$

$$= 2\left[-\frac{x^2e^{-2x}}{2} - \frac{xe^{-2x}}{2} - \frac{e^{-2x}}{4}\right] - e^{-x} + 2\left[-xe^{-x} - e^{-x}\right] + 4e^x$$

$$= -x^2e^{-2x} - xe^{-2x} - \frac{e^{-2x}}{2} - e^{-x} - 2xe^{-x} - 2e^{-x} + 4e^x$$

$$= -x^2e^{-2x} - xe^{-2x} - \frac{e^{-2x}}{2} - 2xe^{-x} - 3e^{-x} + 4e^x$$

Now

$$y_p = -y_1\int \frac{r y_2}{W}\,dx + y_2\int \frac{r y_1}{W}\,dx$$

$$= -e^x\left[-2x^2e^{-x} - 4xe^{-x} - 4e^{-x} + x + x^2 + 2e^{2x}\right]$$

$$+ e^{2x}\left[-x^2e^{-2x} - xe^{-2x} - \frac{e^{-2x}}{2} - 2xe^{-x} - 3e^{-x} + 4e^x\right]$$

$$= 2x^2 + 4x + 4 - xe^x - x^2e^x - 2e^{3x}$$
$$- x^2 - x - \frac{1}{2} - 2xe^x \qquad - 3e^x + 4e^{3x}$$

$$y_p = x^2 + 3x + \frac{7}{2} - 3xe^x - x^2e^x - 3e^x + 2e^{3x}$$

y_p is unique upto a solution of the homogeneous equation. There is $-3e^x$ which will be absorbed in the constant of the general solution. $(c_1 - 3)$

Q-) If the differential eq^{n}

(4) $\frac{dy}{dx} = \sqrt{x^2+y^2}$ $\quad y(1) = 2$

is solved using the Euler's method with step size $h = 0.1$ the find the value of $y(1.2)$ correct to two decimal places

Soln) here $f(x,y) = \sqrt{x^2+y^2}$

$x_0 = 1, y_0 = 2 \quad h = 0.1$

Euler method is given by iterations

$$y_{n+1} = y_n + h f(x_n, y_n)$$

Let $n = 0$

$x_1 = x_0 + h = 1 + 0.1 = 1.1$

$y_1(x_1) = y_0(x_0) + 0.1 f(x_0, y_0)$

$y_1(1.1) = 2 + 0.1\sqrt{(1)^2+(2)^2}$

$= 2 + 0.1\sqrt{5}$

$= 2 + 0.1(2.2360)$

$y_1(1.1) = 2.22360$

Let $n = 1$

$x_2 = x_0 + 2h = 1 + 2 \times 0.1 = 1.2$

$y_2(x_2) = y_1(x_1) + (0.1) f(x_1, y_1)$

$= 2.22360 + (0.1)\sqrt{(1.1)^2 + (2.22360)^2}$

$= 2.22360 + (0.1)\sqrt{6.15439696}$

$= 2.22360 + 0.248079$

$= 2.4711679$

$y(1.2) \approx 2.47$

Solution) 5)

$(\sin^2 x)y'' - (2\sin x \cdot \cos x)y' + (\cos^2 x + 1)y = \sin^3 x$

after rearranging terms

$$y'' - \frac{2\cos x}{\sin x} y' + \frac{(\cos^2 x + 1)}{\sin^2 x} y = \sin x \quad -(1)$$

homogeneous part

$$y'' - \frac{2\cos x}{\sin x} y' + \frac{(\cos^2 x + 1)}{\sin^2 x} y = 0 \quad -(2)$$

To prove y_1 is a solution of homogeneous part.

$y_1 = \sin x$

$y_1' = \cos x \qquad y_1'' = -\sin x.$

$$= -\sin x - 2\frac{\cos x}{\sin x} \cdot \cos x + \frac{(\cos^2 x + 1)}{\sin^2 x} \cdot \sin x$$

$$= -\sin x - \frac{2\cos^2 x}{\sin x} + \frac{\cos^2 x}{\sin x} + \frac{1}{\sin x}$$

$$= -\sin x - \frac{\cos^2 x}{\sin x} + \frac{1}{\sin x} = -\sin x - \frac{1}{\sin x}(\cos^2 x - 1)$$

$$= -\sin x + \sin x = 0$$

So y_1 is solution of (2)

now $y_2 = x\sin x$

$y_2' = \sin x + x\cos x$

$y_2'' = \cos x + (-x\sin x) + \cos x$

$$\cos x - x\sin x + \cos x - 2\frac{\cos x}{\sin x}(\sin x + x\cos x)$$

$$+\frac{(\cos^2 x+1)}{\sin^2 x}\cdot x\sin x$$

$$=\cancel{\cos x}-x\sin x\ \cancel{+\cos x}-\cancel{2\cos x}-2x\frac{\cos^2 x}{\sin x}$$

$$+x\frac{\cos^2 x}{\sin x}+x\frac{1}{\sin x}$$

$$=-x\sin x-x\frac{\cos^2 x}{\sin x}+\frac{x}{\sin x}$$

$$=-x\sin x-\frac{x}{\sin x}(\cos^2 x-1)$$

$$=-x\sin x+\frac{x\sin^2 x}{\sin x}=-x\sin x+x\sin x$$

$$=0$$

So y_2 is soln of (2)

Wronskian

$$W(y_1,y_2)=\begin{vmatrix}\sin x & x\sin x\\ \cos x & \sin x+x\cos x\end{vmatrix}$$

$$=\sin^2 x+x\sin x\cdot\cos x-x\sin x\cos x$$

$$=\sin^2 x$$

on $x\in\left[\frac{\pi}{4},\frac{3\pi}{4}\right]$

$$=\sin^2 x>0$$

So $W(y_1,y_2)=\sin^2 x>0 \quad \forall\, x\in\left[\frac{\pi}{4},\frac{3\pi}{4}\right]$

$\Rightarrow y_1$ are y_2 are l.I on this interval.

$$A = -\int \frac{y_2 R}{W} dx = -\int \frac{x \sin x \cdot \sin x}{\sin^2 x} dx$$

$$= -\int x \, dx = -\frac{x^2}{2}$$

$$B = \int \frac{y_1 R}{W} dx = \int \frac{\sin x \cdot \sin x}{\sin^2 x} dx$$

$$= \int dx = x$$

General solution

$$y(x) = c_1 y_1 + c_2 y_2 + A y_1 + B y_2$$

$$y = c_1 \sin x + c_2 x \sin x - \frac{x^2}{2} \sin x$$

$$+ x^2 \sin x = c_1 \sin x + c_2 x \sin x + x^2/2 \sin x$$

where c_1 & c_2 are arbitrary constants.

Comments about (5)

They can also prove linear independence of y_1 and y_2 by showing

$$\frac{y_1}{y_2} = \frac{\sin x}{x \sin x} = \frac{1}{x} \neq 0 \quad \text{on} \quad \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

$$\left(\text{or} \quad \frac{y_2}{y_1} = \frac{x \sin x}{\sin x} = x \neq 0 \quad \text{on} \quad \left[\pi/4, \frac{3\pi}{4}\right]\right)$$

Q-6 $\left|\frac{dy}{dx}\right| + |y| + a = 0$

for any real valued function $y(x)$

$\left|\frac{dy}{dx}\right| \geq 0$ and $|y| \geq 0$

Case 1: $a > 0$

$\therefore \left|\frac{dy}{dx}\right| + |y| + a \geq a > 0$

1.5

$\Rightarrow \left|\frac{dy}{dx}\right| + |y| + a \neq 0$

No real solution exist when $a > 0$

Case 2: $a = 0$

$\left|\frac{dy}{dx}\right| + |y| = 0$

this is possible only when

$\frac{dy}{dx} = 0$ and $y = 0$

1

$y(x) \equiv 0$ is the solution of diff eq^n for $a = 0$

$\therefore$ A real solution exists when $a = 0$.

Case 3

Let $a < 0$.

Then consider the function $y(x) = +a$ (or $y(x) = -a$) Then $\frac{dy}{dx} = 0$

Then $|y(x)| = -a$ (since $a < 0$, $|a| = -a$)

$|a| = -a$

1.5

Now, $\left|\frac{dy}{dx}\right| + |y| + a$

$= 0 + (-a) + a = 0$

(or $y(x) = -a$)

So, $y(x) = a$ is a real solution.

Thus the given Differential Equation has real solution iff $a \leq 0$

½ (1) mark